Read → Images   CV2.imread('PathTo image')
→ Videos   CV2.VideoCapTure('Path to video')

Resizing & Rescalling
→ w = int(img.shape[1] * scale)
→ h = int(img.shape[0] * scale)
→ d = (w, h)
→ cv2.resize(img, d, interpolation = cv2.INTER_AREA

Draw & Put Text

Rect → CV2.rectangle(blank, (x1, y1), (x2, y2), (0, 255, 0), Thickness = 2)
Circle → CV2.circle(blank, (c_x, c_y), r, (0,0,255))
Line → CV2.Line(blank, (x1, y1), (x2, y2), color, Thickness)
Text → CV2.PutText(blank, 'Text', (x, y), Font, scale, color, Thickness)

Basic func

1) Gray → CV2.cvtColor(img, CV2.COLOR_BGR2GRAY)
2) Blur → CV2.GaussianBlur(img, (K, K), 0)
3) Edge detection → CV2.Canny(img, t1, t2)
4) Cropping → cropped = img[y1:y2, x1:x2]

Transformation → CV2.wrapAffine(x, y)

Left -x
Right +x
up -y
down +y

Contours → Similar to edges but mathematically different

Find → cnts, h = cv2.findContours(canny_edges, mode, Approx)

mode → RETR_LIST (all)
→ RETR_EXTERNAL (only outside ones)
→ RETR_TREE (hierarichial)

Approx → Chain_Approx_None (all Points)
→ Chain_APPROX_SIMPLE (compresses lines into end-points)

Draw → cv2.drawContours(blank, contours, -1, (0,0,255), 1)

-1 → draw all

Color spaces & Channels

OpenCV → BGR

Matplotlib → RGB

split → b,g,r = cv2.split(img

merge → img = cv2.merge([b,g,r])

BitWise & Masking

AND → Intersecting regions

OR → intersecting & Non-intersecting

XOR → Non-Intersecting

NOT → invert

masking → Focus on specific Area using cv2.bitwise_and(img, mask)